

Towards Energy-Aware Design Choices in Event-Driven Microservices

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As software systems increasingly transition toward cloud-native microservices and event-driven architectures (EDA), decisions regarding service topology, message broker selection, and data serialization formats are traditionally guided by scalability, resilience, and responsiveness. However, these structural choices profoundly influence CPU cycles, network traffic, memory pressure, and message broker overhead—metrics that directly translate into physical hardware energy consumption. For organizations deploying cloud-native distributed systems at scale, minor inefficiencies in per-event processing compound into massive environmental and infrastructure costs.

The motivation for this research is to evaluate software design patterns for microservices through a rigorous sustainability lens, recognizing that macro-level architectural choices often exert a more significant long-term energy impact than micro-level code optimizations. This project proposes an empirical study to determine whether common event-driven design decisions can be systematically ranked by their energy consumption per business event, enabling architects to make greener choices without sacrificing system quality.

The experimental scope focuses on a controlled microservice testbed (4–6 services) deployed via containers on a standardized machine. The study evaluates three critical, practical dimensions:

1. **EDA Topology:** Chain versus Parallel Fan-Out. Current empirical research indicates that dense Chain topologies often experience steep increases in energy consumption at scale, whereas Parallel Fan-Out configurations tend to emerge as highly energy-efficient under CPU-bound loads.
2. **Broker Choice:** Apache Kafka versus RabbitMQ. The study will evaluate the energy overhead of Kafka's distributed log architecture, which guarantees high throughput, against RabbitMQ's traditional message broker approach with durable queues.
3. **Serialization Format:** JSON versus Protobuf. Binary formats like Protobuf typically produce messages that are 3 to 5 times smaller than JSON, potentially reducing network transmission energy and computational deserialization overhead.

The workload will simulate realistic asynchronous business workflows, such as order processing, user notifications, and inventory updates. For each configuration, the system will be stress-tested under controlled event rates. Measured metrics will include latency per event, system throughput, CPU and memory utilization, and network bytes transferred. Crucially, the study will map these metrics to the standardized Software Carbon Intensity (SCI) specification, calculating the precise rate of energy and carbon emissions relative to a functional unit (e.g., per 1,000 events).

The expected contribution of this research is a highly empirical decision guide for sustainable event-driven software design, identifying the optimal combinations of topology, broker, and serialization to maximize energy efficiency.